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MySpot

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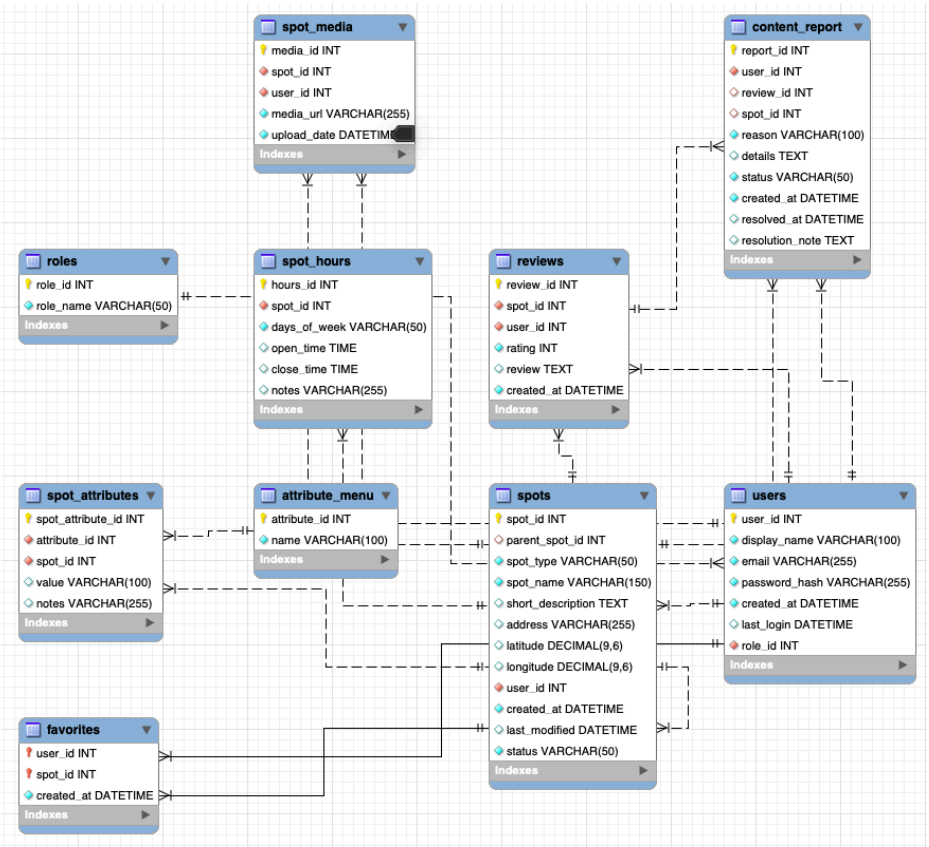
1. Brief Introduction

The MySpot system is a database-driven web application designed to help students discover, evaluate, and share study locations throughout the Blacksburg and Virginia Tech area. Its primary goal is to centralize information about study spots, including amenities, availability, reviews, ratings, and media, so that users can make informed decisions based on their individual preferences. By organizing this data within a relational database, the system ensures consistency, efficient querying, and reliable management of both location data and user-generated content.

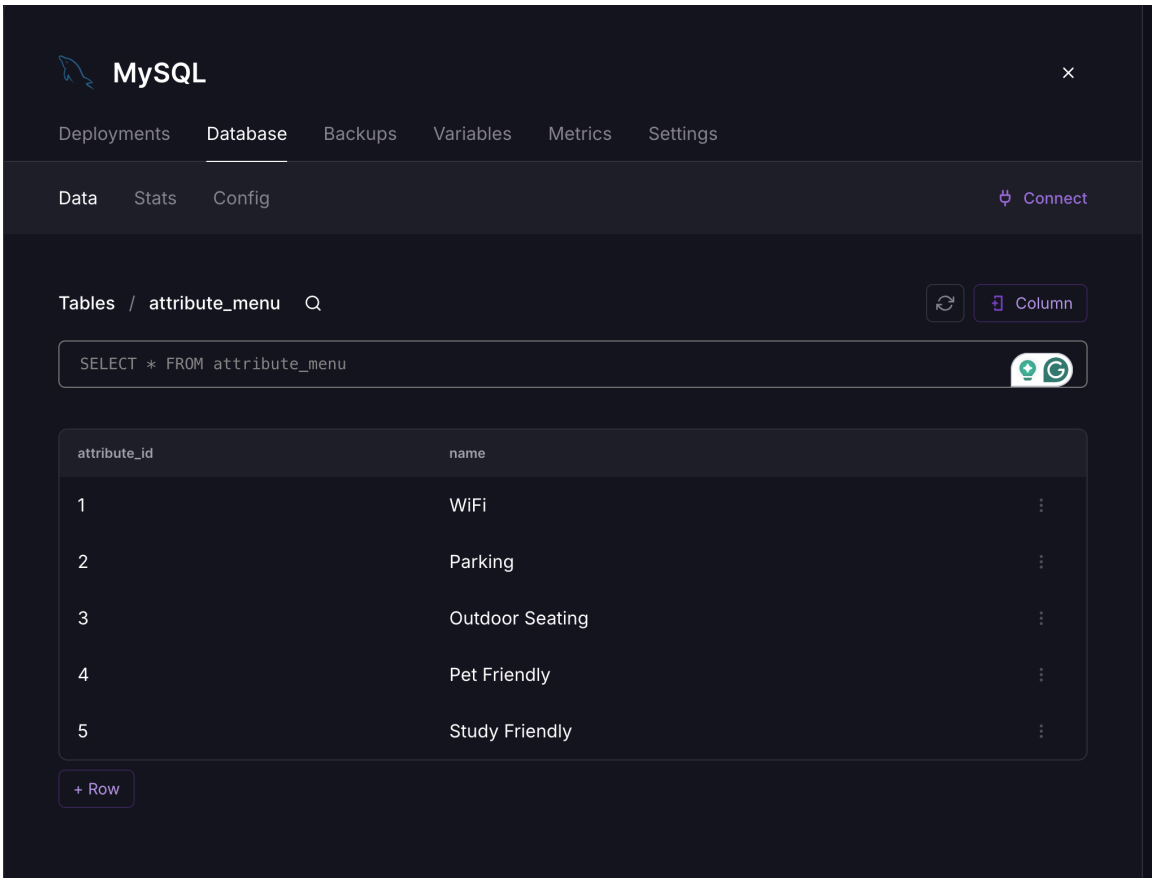
The system supports multiple target scenarios involving different types of users. Students actively use the platform to browse study spots, filter locations based on attributes such as noise level or available amenities, and contribute content through reviews, ratings, and new spot submissions. Administrators interact with the system to moderate content, manage users, and maintain data integrity so the platform remains accurate and trustworthy. In addition, prospective students or visitors can use the application in a read-only mode to explore study environments and gain insights into campus life. Overall, MySpot supports both interactive and exploratory use cases while addressing the lack of a centralized and structured solution for finding suitable study locations.

2. Normalized Relational Schema



3. Screenshots of Tables with Sample Data

attribute_menu



content_report

MySQL

Deployments Database Backups Variables Metrics Settings

Data Stats Config Connect

Tables / content_report Column

```
SELECT * FROM attribute_menu
```

report_id	user_id	review_id	spot_id	reason	details	status
1	1	2	NULL	Spam	This review seems overly repetitive and possib...	open
2	2	NULL	4	Incorrect Info	The listed hours do not really apply because th...	open
3	3	5	NULL	Off Topic	Review talks more about class schedule than t...	resol
4	4	NULL	2	Duplicate	This spot may have been entered twice by mis...	open
5	5	1	NULL	Other	Just wanted moderator review on whether this...	open

+ Row

favorites

MySQL

Deployments | Database | Backups | Variables | Metrics | Settings

Data | Stats | Config | Connect

Tables / favorites

Column

```
SELECT * FROM attribute_menu
```

user_id	spot_id	created_at
1	3	2026-03-19 21:33:28
2	1	2026-03-19 21:33:28
3	5	2026-03-19 21:33:28
4	4	2026-03-19 21:33:28
5	2	2026-03-19 21:33:28

+ Row

reviews

MySQL

Deployments | Database | Backups | Variables | Metrics | Settings

Data | Stats | Config | Connect

Tables / reviews

Column

```
SELECT * FROM attribute_menu
```

review_id	spot_id	user_id	rating	review	created_at
1	1	2	5	Really good coffee and a solid place to sit and ...	2026-03-19 21:33:28
2	2	1	4	Great pizza especially late at night, but it can g...	2026-03-19 21:33:28
3	3	4	5	Best study spot on campus with lots of space ...	2026-03-19 21:33:28
4	4	5	4	Nice place to hang out when the weather is go...	2026-03-19 21:33:28
5	5	3	4	Convenient on-campus coffee stop and usuall...	2026-03-19 21:33:28

+ Row

spot_hours

Tables / spot_hours

Column

```
SELECT * FROM attribute_menu
```

hours_id	spot_id	days_of_week	open_time	close_time	notes
1	1	Mon-Fri	07:00:00	19:00:00	Hours may vary during breaks
2	2	Daily	11:00:00	23:59:00	Popular late-night option
3	3	Daily	07:00:00	23:00:00	Can vary during exam periods
4	4	Daily	NULL	NULL	Open outdoor public space
5	5	Mon-Fri	08:00:00	17:00:00	Mostly busiest during class hours

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roles

MySQL

Deployments Database Backups Variables Metrics Settings

Data Stats Config

Connect

Tables / roles

Column

```
SELECT * FROM attribute_menu
```

role_id	role_name
1	admin
5	guest
3	moderator
4	owner
2	user

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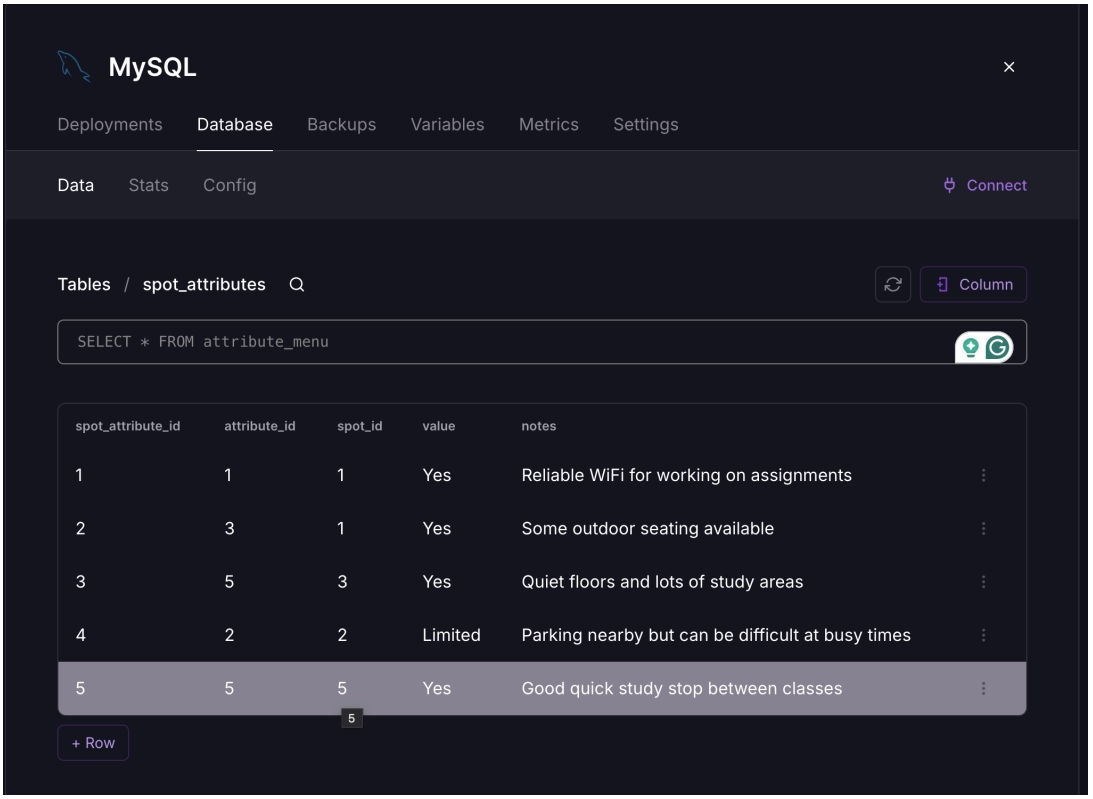
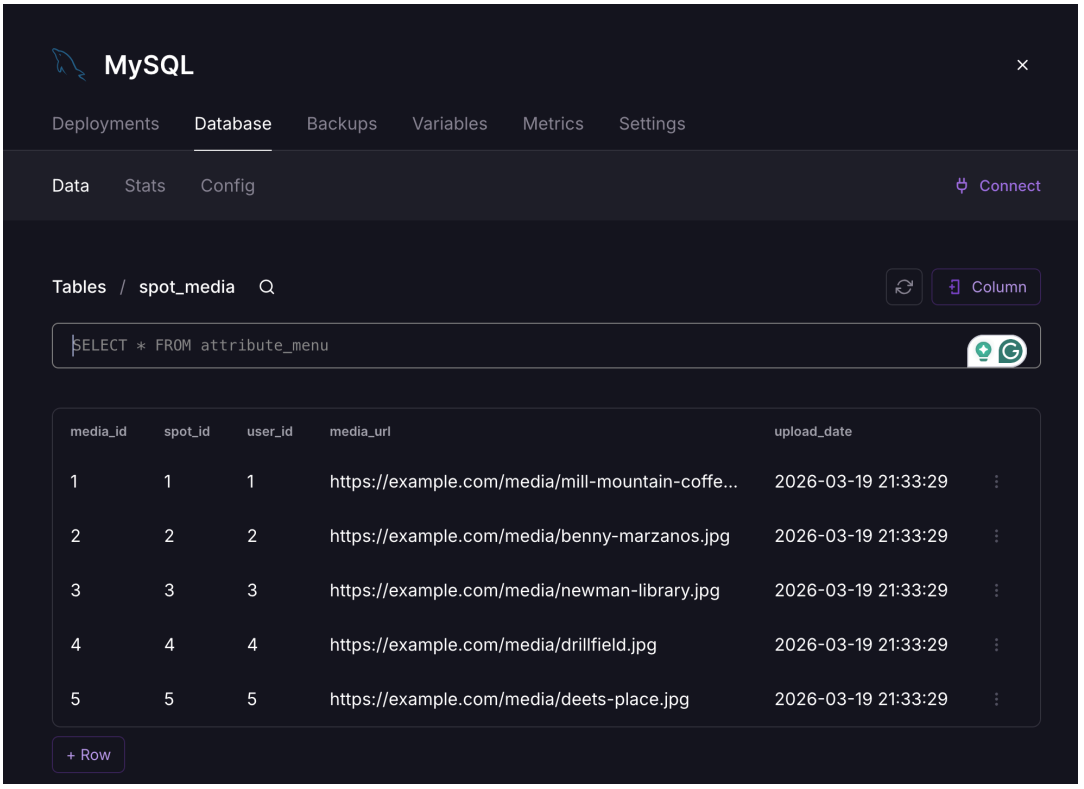
spot_attributes



spot_attribute_id	attribute_id	spot_id	value	notes
1	1	1	Yes	Reliable WiFi for working on assignments
2	3	1	Yes	Some outdoor seating available
3	5	3	Yes	Quiet floors and lots of study areas
4	2	2	Limited	Parking nearby but can be difficult at busy times
5	5	5	Yes	Good quick study stop between classes

spot_media



media_id	spot_id	user_id	media_url	upload_date
1	1	1	https://example.com/media/mill-mountain-coffe...	2026-03-19 21:33:29
2	2	2	https://example.com/media/benny-marzanos.jpg	2026-03-19 21:33:29
3	3	3	https://example.com/media/newman-library.jpg	2026-03-19 21:33:29
4	4	4	https://example.com/media/drillfield.jpg	2026-03-19 21:33:29
5	5	5	https://example.com/media/deets-place.jpg	2026-03-19 21:33:29

spots

MySQL

Deployments | Database | Backups | Variables | Metrics | Settings

Data | Stats | Config | Connect

Tables / spots

Column

```
SELECT * FROM attribute_menu
```

spot_id	parent_spot_id	spot_type	spot_name	short_description
1	NULL	Coffee Shop	Mill Mountain Coffee - Blacksburg	Popular coffee spot for studying an
2	NULL	Restaurant	Benny Marzano's	Late-night pizza favorite near camp
3	NULL	Library	Newman Library	Main Virginia Tech library with lots
4	NULL	Outdoor Space	Drillfield	Open campus green space in the c
5	NULL	Cafe	Deet's Place	Campus coffee shop good for quicl

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users

MySQL

Deployments | Database | Backups | Variables | Metrics | Settings

Data | Stats | Config | Connect

Tables / users

Column

```
SELECT * FROM attribute_menu
```

user_id	display_name	email	password_hash	created_at	last_login	role_id
1	Liam Erickson	liam@vt.edu	hash1	2026-03-19 21:33:28		1
2	Sarah Collins	sarah@vt.edu	hash2	2026-03-19 21:33:28		2
3	Marcus Lee	marcus@vt.edu	hash3	2026-03-19 21:33:28		3
4	Emma Carter	emma@vt.edu	hash4	2026-03-19 21:33:28		2
5	Noah Bennett	noah@vt.edu	hash5	2026-03-19 21:33:28		4

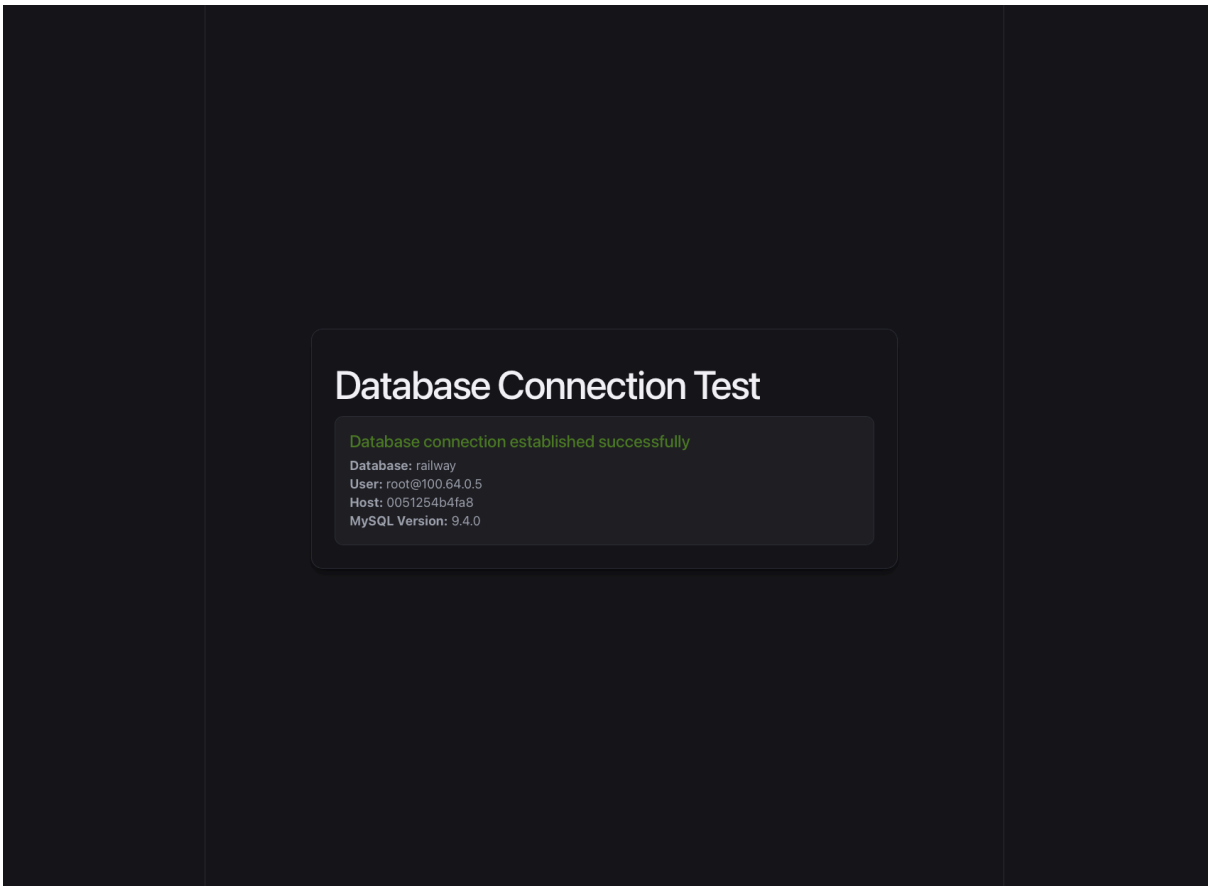
+ Row

4. Confirmation of DBMS-Interface Connection

1. DBMS Information:

- **The name of the DBMS system used:** MySQL (deployed using Railway)
- **The name of the interface language:** React, Javascript, CSS
- The application is hosted on Vercel

2. Submit a screenshot confirming successful connection establishment between the DBMS and the interface.



The same is available on the application hosted at https://my-spot-omega.vercel.app/